

LD0-K Canonical Discrete Density-Kernel Specification

One periodized node stencil for exact dense direct and FFT convolution

mdstats development specification

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Status and scope

This specification implements architecture gate **LD0-K** for `mdstats 0.19.43a0`. It is subordinate to the *Dynamical Framework and Atomic Density Architecture Standard*.

The gate introduces the canonical smoothing operator `discrete_periodized_v1` alongside the unchanged compatibility operator `legacy_spectral_v1`.

This gate changes no default. Existing calls continue to use `legacy_spectral_v1` unless the canonical operator is selected explicitly.

This gate does not implement:

- the effective CIC-plus-stencil broadening policy owned by LD0-B;
- local-sparse storage or sparse convolution;
- automatic backend selection;
- framework-edge quadrature refinement;
- mesh or HDR algorithm changes.

Motivation

The previous dense estimator deposits weighted samples by periodic trilinear cloud-in-cell assignment and smooths the node masses with a finite-mode spectral Gaussian multiplier. A future sparse backend cannot apply that globally supported finite spectral operator locally without approximation.

LD0-K therefore defines one finite-support discrete periodized node stencil. The dense reference backend can apply this stencil either by direct circular convolution or by FFT circular convolution. A later sparse backend will scatter the same stored weights from occupied CIC nodes. The estimator boundary is then explicit and testable.

Scientific operator

Let the row-vector display cell be

$$H = \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \mathbf{a}_3 \end{bmatrix},$$

and let the logical grid shape be

$$\mathbf{N} = (N_1, N_2, N_3).$$

For an integer grid-image vector $\mathbf{q} \in \mathbb{Z}^3$, define

$$\Delta \mathbf{x}_{\mathbf{q}} = \left(\frac{q_1}{N_1}, \frac{q_2}{N_2}, \frac{q_3}{N_3} \right) H.$$

The voxel volume is

$$\Delta V = \frac{|\det H|}{N_1 N_2 N_3}.$$

For $\sigma > 0$, the unnormalized retained contribution is

$$a_{\mathbf{q}} = \Delta V \frac{1}{(2\pi\sigma^2)^{3/2}} \exp \left(-\frac{\|\Delta \mathbf{x}_{\mathbf{q}}\|_2^2}{2\sigma^2} \right) \left[\|\Delta \mathbf{x}_{\mathbf{q}}\|_2 \leq r_{\text{cut}} \right].$$

The cutoff is selected from the radial mass of a three-dimensional isotropic Gaussian. Since $r^2/\sigma^2 \sim \chi_3^2$,

$$r_{\text{cut}} = \sigma \sqrt{F_{\chi_3^2}^{-1}(1 - \varepsilon)},$$

where

$$10^{-15} \leq \varepsilon \leq 10^{-3}.$$

Each $\mathbf{q}$ is mapped to the canonical periodic node offset

$$\boldsymbol{\delta} = \mathbf{q} \bmod \mathbf{N}.$$

All retained image contributions mapping to the same canonical offset are summed:

$$A_{\boldsymbol{\delta}} = \sum_{\mathbf{q} \equiv \boldsymbol{\delta} \pmod{\mathbf{N}}} a_{\mathbf{q}}.$$

With

$$S_A = \sum_{\boldsymbol{\delta}} A_{\boldsymbol{\delta}},$$

the canonical stencil is

$$g_{\boldsymbol{\delta}} = \frac{A_{\boldsymbol{\delta}}}{S_A}.$$

Thus

$$g_{\delta} \geq 0, \quad \sum_{\delta} g_{\delta} = 1.$$

For deposited node masses m_j , the smoothed node masses are

$$\tilde{m}_g = \sum_j m_j g_{g-j}.$$

The density is

$$\rho_g = \frac{\tilde{m}_g}{\Delta V}.$$

For $\sigma = 0$, the operator is exactly the identity:

$$g_0 = 1, \quad g_{\delta \neq 0} = 0.$$

Conservative support enumeration

The implementation enumerates integer vectors $\mathbf{q}$, not separate canonical offsets and image vectors. This guarantees that every retained periodic image contribution is counted exactly once.

From

$$\mathbf{f} = \Delta \mathbf{x} H^{-1},$$

and $\| \Delta \mathbf{x} \|_2 \leq r_{\text{cut}}$, a conservative coordinate bound is

$$|f_i| \leq r_{\text{cut}} \| H_{:,i}^{-1} \|_2.$$

Therefore

$$|q_i| \leq B_i = \left\lceil N_i r_{\text{cut}} \| H_{:,i}^{-1} \|_2 \right\rceil.$$

The rectangular integer box $[-B_1, B_1] \times [-B_2, B_2] \times [-B_3, B_3]$ is scanned in deterministic lexicographic order and filtered by the exact Cartesian metric. Enumeration is chunked so that temporary arrays are bounded independently of the total candidate count.

Data structures

PeriodicGaussianStencil

```
@dataclass(frozen=True, slots=True)
class PeriodicGaussianStencil:
    grid_shape: tuple[int, int, int]
    display_cell: NDArray[np.float64]
    gaussian_bandwidth: float
    kernel_tail_tolerance: float
    cutoff_radius: float
    values: NDArray[np.float64]
    active_flat_indices: NDArray[np.int64]
    active_weights: NDArray[np.float64]
    pre_normalization_sum: float
    normalization_factor: float
    periodic_image_contribution_count: int
    covariance: NDArray[np.float64]
    metadata: FrozenJSONMapping
    schema_version: str
```

All arrays are C-contiguous, defensive, read-only copies. `values` has exactly the logical grid shape. `active_flat_indices` are strictly increasing C-order flat indices of positive canonical stencil entries. `active_weights` are aligned with those indices.

The covariance is computed from the unaggregated retained image contributions:

$$C_g = \frac{1}{S_A} \sum_{\mathbf{q}} a_{\mathbf{q}} \Delta \mathbf{x}_{\mathbf{q}} \Delta \mathbf{x}_{\mathbf{q}}^T.$$

It is stored for LD0-B but does not yet alter resolution selection.

Diagnostics

Every canonical field records:

```
smoothing_operator = discrete_periodized_v1
kernel_tail_tolerance
continuous_tail_mass_bound
kernel_cutoff_radius
stencil_pre_normalization_sum
stencil_normalization_factor
stencil_offset_count
periodic_image_contribution_count
stencil_covariance_cartesian
canonical_convolution_method = fft
canonical_negative_roundoff_clipped
canonical_post_convolution_normalization_factor
```

1-S_A is not called an omitted discrete mass because it combines continuous-tail truncation and sampling quadrature error.

Public numerical functions

```
build_periodic_gaussian_stencil(
    grid_shape,
    display_cell,
    gaussian_bandwidth,
    *,
    kernel_tail_tolerance=1e-8,
) -> PeriodicGaussianStencil

convolve_periodic_stencil_direct(
    mass_grid,
    stencil,
) -> NDArray[np.float64]

convolve_periodic_stencil_fft(
    mass_grid,
    stencil,
) -> NDArray[np.float64]

smooth_periodic_node_masses(
    mass_grid,
    display_cell,
    gaussian_bandwidth,
    kernel_options,
) -> tuple[NDArray[np.float64], FrozenJSONMapping]
```

The direct implementation is a correctness oracle. Production dense canonical fields use FFT circular convolution.

Floating-point completion policy

The canonical stencil is nonnegative and normalized before convolution.

After FFT convolution:

1. define

$$\tau_- = 64\epsilon_{\text{mach}} \max\left(1, \max_g |\tilde{m}_g|\right);$$

2. reject values below $-\tau_-$ as a numerical failure;

3. set only negative values in $[-\tau_-, 0)$ to zero;
4. do not clip positive tails;
5. renormalize the smoothed node masses to the original deposited mass.

Direct convolution passes through the same completion function. This keeps direct and FFT comparisons under one declared policy while preserving all positive canonical stencil mass.

The legacy operator keeps its historical clipping and renormalization behavior unchanged.

Operator selection and compatibility

Backend	legacy_spectral_v1	discrete_periodized_v1
dense	supported and default	supported explicitly
local_sparse	rejected	reserved for LD1-B
auto	rejected before LD4	rejected before LD4

The default operator remains `legacy_spectral_v1` in this release.

Planning integration

For a canonical dense field with $\sigma > 0$:

```
stencil_value_count = logical_node_count
```

because production FFT convolution materializes one dense stencil array. For the identity path $\sigma = 0$:

```
stencil_value_count = 0
```

Phase A uses the logical node count as the canonical stencil upper bound. Phase C checks the scene-wide sum against `max_density_stencil_values`.

Planning metadata records the requested operator for every field and the set of operators present in the approved scene. Mixed atomic/framework operator choices are permitted because every field is scientifically self-describing.

Inputs and constraints

- `grid_shape` contains three positive integers.
- `display_cell` is finite, shape (3, 3), and nonsingular.
- `gaussian_bandwidth` is finite and nonnegative.
- `kernel_tail_tolerance` lies in $[1e-15, 1e-3]$.
- `mass_grid` is finite, nonnegative within numerical tolerance, and has the same shape as the stencil.
- Full three-dimensional periodicity is required by the current density backend.

Determinism

For fixed inputs:

- support bounds are deterministic;
- integer vectors are enumerated lexicographically;
- canonical aggregation is deterministic;
- active indices are C-order sorted;
- direct accumulation order is fixed;
- metadata field ordering is canonical;
- repeated builds are byte-identical on one NumPy/SciPy platform.

Error handling

Raise `GraphStyleError` for invalid user kernel parameters.

Raise `GraphAdapterError` for malformed cells, shapes, arrays, zero normalization, or FFT negativity beyond the declared roundoff tolerance.

Raise `GraphComplexityError` if the conservative support-enumeration candidate count exceeds the implementation safety bound before scanning begins.

Required focused tests

Stencil construction

1. identity stencil for $\sigma=0$;
2. nonnegative normalized stencil;
3. deterministic repeat build;
4. orthorhombic and skewed cells;
5. a support radius spanning multiple periodic images;
6. monotone cutoff growth as tail tolerance decreases;
7. exact active-index/value alignment;
8. covariance symmetry and positive semidefiniteness within tolerance.

Convolution

For orthorhombic, LTA-primitive, random positive mass, CIC mass, and multiple-image support cases:

```
direct-vs-FFT relative L1 <= 5e-12
direct-vs-FFT relative L-infinity <= 2e-11
stencil sum absolute error <= 5e-15
integral error <= 5e-13 * max(1, total_measure)
```

Integration

- atomic canonical fields integrate to selected occupancy;
- framework-vertex canonical fields integrate to projected vertex count;
- framework-edge canonical fields integrate to mean retained arc length;
- integer periodic translations are invariant;
- canonical metadata and planning counts are present;
- `legacy_spectral_v1` remains the default and numerically identical to `mdstats 0.19.42a0` fixtures;
- canonical and legacy operators may differ on under-resolved inputs without either being mislabeled.

Acceptance gate

LD0-K passes only if:

1. all focused tests satisfy the fixed tolerances above;
2. the legacy default path is exactly preserved on matched fixtures;
3. `discrete_periodized_v1` works for atomic, framework-vertex, and framework-edge dense fields;
4. the zero-bandwidth path allocates no Gaussian stencil and returns exact CIC masses;
5. planning and realized metadata identify the operator consistently;
6. no default operator migration occurs;
7. documentation and code distinguish standard Fourier convolution and Gaussian probability background from the project-specific canonical stencil policy.

Borrowed, standard, and project-specific material

- **Borrowed algorithm:** periodic trilinear cloud-in-cell assignment follows Hockney and Eastwood.
- **Standard mathematical background:** FFT circular convolution, the isotropic Gaussian density, and the χ_3^2 radial law.
- **Project-specific design:** finite Cartesian support, canonical image aggregation, exact dense/sparse operator identity, diagnostics, roundoff completion, migration policy, and gate tolerances.

References

1. R. W. Hockney and J. W. Eastwood, *Computer Simulation Using Particles*, Adam Hilger, 1988. Reprint DOI: 10.1201/9780367806934.